
Foresight™ Chromatography Columns, 1 and 5 mL

Instruction Manual



BIO-RAD

Foresight Chromatography Columns, 1 and 5 mL

Instruction Manual

Foresight 1 mL Columns Catalog Numbers

732-4720
732-4721
732-4722
732-4723
12007020
12004038
12007392
732-4730
732-4732
732-4735
732-4736
12003150
12020456

Foresight 5 mL Columns Catalog Numbers

732-4740
732-4741
732-4742
732-4743
12007021
12004037
12007393
732-4750
732-4752
732-4755
732-4756
12003149
12020484



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Nuvia aPrime 4A Media is covered by U.S. Patent Numbers 9,669,402 and 10,682,640, and foreign counterparts. Nuvia cPrime Media is covered by U.S. Patent Numbers 9,309,282, 10,792,654, and 11,590,486, and foreign counterparts. Nuvia wPrime 2A Media is covered by U.S. Patent Numbers 9,669,402, 10,682,640, and 11,896,969, and foreign counterparts.

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Revision History

Document	Date	Description of Change
Foresight Chromatography Media Columns, 1 and 5 mL Bulletin No. 10024831 Rev C	June 2025	Reformat and edit, and add catalog numbers
Foresight Chromatography Media Columns, 1 and 5 mL Bulletin No. 10024831 Rev B	2013	Add catalog numbers
Foresight Chromatography Media Columns, 1 and 5 mL Bulletin No. 10024831 Rev A	2012	New document

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Chapter 1 Introduction

Foresight™ chromatography media columns feature a robust design process that scientists can use for qualitative media screening experiments and/or small scale methods development. Utilizing Foresight columns through the design of experiments (DOE) process, an operational window can be defined to aid in developing a quality by design (QbD) process.

Foresight columns allow faster, more convenient experimental setup, and the design offers reliability and convenience. Each column is individually packed under optimum compression, ensuring consistent experimental results and leak-free operation at pressures up to 30 bar (consult the resin manuals for pressure limit criteria). Prepacked columns are ready for equilibration in the buffer of choice, and bed heights are available in ratios that allow straightforward comparative evaluation.

Foresight columns are available in a variety of chromatography modes, including anion, cation, and mixed-mode. For information on the complete Foresight column product line, see [Appendix C, Ordering Information](#).

Column Properties and Specifications

Table 1. Foresight chromatography media column properties

Property	Description
Connectors	10–32 fittings
Column material	Polyethylene (HDPE)
Frit material	Polypropylene/polyethylene
Buffer shipping conditions:	
■ UNOsphere IEX media, Nuvia™ IEX media	20% EtOH plus 150 mM NaCl
■ Ceramic Hydroxyapatite Media (CHT)	20% EtOH + 10 mM sodium phosphate, pH 6.8
■ Nuvia Mixed-Mode Media, Nuvia IMAC Media	20% EtOH
Autoclavability	Not autoclavable

Table 2. Foresight column dimensions and operating specifications

Specification	1 mL	5 mL
Dimensions:		
■ Inner diameter (ID)	8 mm	8 mm
■ Length	20 mm	100 mm
Maximum pressure	20 bar (290 psi)	30 bar (435 psi)
Recommended flow rates*	■ 0.5–1 mL/min ■ 59.5–119 cm/hr	■ 2.5–5 mL/min ■ 297.5–595 cm/hr
Maximum flow rate*	■ 8 mL/min ■ 952 cm/hr	■ 8 mL/min ■ 952 cm/hr

*Refer to the chromatography resin manual for resin-specific flow rate properties.

Chapter 2 Connecting to Low Pressure and Liquid Chromatography Systems

Foresight columns have standard HPLC connections (M10–32 UNF for 1/16" tubing), and are therefore compatible with low-pressure and liquid chromatography systems. Review the best practices in the following section, and then complete the steps in [Connecting to Low Pressure and Liquid Chromatography Systems on page 9](#).

Best Practices for Preparing the Column Tubing

Table 3. Best practices

System	Recommendations
Medium to high pressure chromatography systems (NGC Chromatography System)	■ Bio-Rad recommends 1/16" OD inlet and outlet tubing. ■ If the DuoFlow system tubing has male 1/4–28 fittings attached, use male 10–32 to female 1/4–28 adaptors (catalog #750-0564) to connect the column to the DuoFlow system. ■ Alternatively, use long 10–32 fittings (catalog #760-1311) to attach to the tubing.
AKTA systems	■ Bio-Rad recommends 1/16" OD inlet and outlet tubing. ■ If the system tubing has male 1/16" fittings, these attach directly to the column. ■ If tubing fittings are required, use long 10–32 fittings (catalog #760-1311) to attach to the tubing.
Low pressure chromatography systems	■ 0.8 or 1.6 mm tubing is recommended. ■ Use IDEX Health & Science P-642 female luer to male 10–32 adaptor fittings to connect to column. ■ To attach to the respectively-sized tubing on the BioLogic LP system, use either 0.8 mm barb to male luer (catalog #731-8224) 1.6 mm barb to male luer (catalog #731-8225) 3.2 barb to male luer (catalog #731-8226)

Connecting the Column to a System

To connect the column

1. Mount the column in a vertical position with the arrow pointed downward.
2. Remove the stop plugs from the top and bottom of the column.
3. Connect the inlet tubing to the solvent delivery system, and then, with a flow rate of 0.2mL/min, connect the inlet tubing to the column:
 - a. Press the tubing until it is situated firmly against the bottom.
 - b. Tighten the 1/16 in fingertight male fitting to the female end piece on the column.
4. When the solvent is flowing freely from the outlet end of the column, connect the outlet tubing to the column:
 - a. Press the tubing until it is situated firmly against the bottom
 - b. Tighten the 1/16 in fingertight male fitting to the female end piece on the column.
5. Ensure that the solvent is present in the column assembly, and then connect the assembly to the chromatography system detector.

Chapter 3 Preparing a Column for Use

Buffer shipping conditions are specified in [Column Properties and Specifications on page 7](#). All columns are ready to use after equilibrating in the appropriate buffer. For more information, refer to the resin manual.

Complete the following procedure to a) perform a buffer exchange, b) connect the column to a liquid chromatography system or peristaltic pump, and c) condition the column.

To prepare the column

1. Set the pump flow rate as follows, based on resin specifications (for information, refer to the resin manual).
 - For the 1mL column, set to 0.5–1 mL/min (59.5–119 cm/hr)
 - For the 5 mL column, set to 2.5–5 mL/min (297.5–595 cm/hr)
2. Wash the column with appropriate buffers.
3. Equilibrate 5-column volumes with appropriate buffers.

Note: For information on buffers, refer to the resin manual.
4. Reduce the flow rate to the rate that will be used in the purification protocol.

Sample Preparation

Correct pH and ionic strength are necessary to achieve consistent and reproducible results. To exchange sample into the starting buffer, or dilute to the starting buffer concentration, you can do one of the following:

- Dilute the sample to the ionic strength of the starting buffer
- Dialyze against the start buffer
- Exchange into the starting buffer

Depending on your sample volume, you can perform a buffer exchange using the products specified in [Table 4](#). Before column application, filter all samples through a 0.45 µm filter.

Table 4. Products for buffer exchange

Sample volume	Product	MW cutoff	Catalog no.
10–75 µL	Micro Bio-Spin P-6 Column	6 kD	732-6221
	Micro Bio-Spin P-30 Column	30 kD	732-6223
50–100 µL	Bio-Spin P-6 Column	6 kD	732-6227
	Bio-Spin P-30 column	30 kD	732-6231
Up to 1.5 mL	EconoFit Bio-Gel P-6 Desalting Column	6 kD	12009238
Up to 3 mL	Econo-Pac 10DG Desalting Column	6 kD	732-2010
Up to 3 mL	Bio-Scale Mini Bio-Gel P-6 Desalting Cartridges, 10 mL	6 kD	732-5304
Up to 15 mL		6 kD	732-5312
Unlimited	Bio-Gel P-6DG gel	6 kD	150-0738

General Purification Protocol

Recommendations on general purification protocols for each chromatography media are available in the corresponding instruction manual. For more information on each resin, refer to the list in [Appendix A, Chromatography Media Documentation](#).

Important: For best results and increased column life, samples and buffers should be degassed and filtered through a 0.45 µm filter.

Purification Scale-Up

For quick scale-up, you can connect two or three columns of the same type in series. Foresight columns are available in 1 and 5 mL column formats. All chromatography media are designed for process scale and are available in larger bulk quantities from for process development to manufacturing scale. All media are fully supported with regulatory support files.

Important: Backpressure Increases with columns in series, so care should be taken to maintain pressure at ≤ 20 bar for 1ml and ≤ 30 bar for 5 mL columns while considering chromatography resin pressure specifications.

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Chapter 4 Cleaning, Sanitization, and Storage

If a column no longer yields reproducible results, the media might require thorough cleaning-in-place (CIP) and sanitation to remove strongly bound contaminants.

Acceptable Clean-In-Place Agents

For information on CIP agents, see [Chromatography Media Documentation on page 17](#).

Clean-in-Place Procedure

To clean-in-place and sanitize

1. While maintaining a contact time of at least 40 min, sanitize the support in the column with 2–4 bed volumes of appropriate buffer.
2. To equilibrate the column again, wash it with 2–4 bed volumes of the equilibration buffer.

Autoclaving

Important: Foresight columns are not autoclavable.

Storage

After washing the columns with deionized water, Foresight columns should be a) purged, b) stored in the recommended solution for the specific media, and c) capped for extended storage.

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Appendix A Chromatography Media Documentation

Table 5. Chromatography resin and media documentation

Chromatography Media	Mode	Document No.
Nuvia S	Strong Cation	10018215
Nuvia HR-S	Strong Cation	10033316
Nuvia Q	Strong Anion	10018215
Nuvia HP-Q	Strong Anion	10000097131
Nuvia IMAC	Ni-Charged	D116698
Nuvia aPrime 4A	Mixed Mode (Hydrophobic-Anionic)	10000098359
Nuvia cPrime	Mixed Mode (Hydrophobic-Weak Cationic)	10023853
Nuvia wPrime 2A	Mixed Mode (Hydrophobic-Weak Anionic)	10000245002
UNOsphere S	Strong Cation	4110109
UNOsphere Q	Strong Anion	4110109
CHT Type I	Mixed Mode Metal Affinity Cationic	6086
CHT Type II	Mixed Mode Metal Affinity Cationic	6086
CHT XT	Mixed Mode Metal Affinity Cationic	6086

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Appendix B Related Reading

- Harris, E.L.V. and Angal, S. (1989) Protein Purification Methods: A Practical Approach (Oxford: IRL Press)
- Scopes, R.K. (1987) Protein Purification: Principles and Practice, Second Edition (New York: Springer-Verlag)
- Snyder, L.R. and Kirkland, J.J. (1979) Introduction to Modern Liquid Chromatography, Second Edition (New York: Wiley)
- Gagnon, P. (1997) Avoiding Instrument-associated Aberrations in Purification Scale-up and Scale-down, BioPharm 10, 42–45
- Fittings for low pressure systems: <http://www.idex-hs.com/Default.aspx>

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Appendix C Ordering Information

The tables in this section contain ordering information for media columns, media filter plates, RoboColumns, and fittings kits.

Table 6. Foresight Chromatography Media Columns

Product	Mode	Catalog No.
Foresight Nuvia S Column 1 mL	Strong Cation	732-4720
Foresight Nuvia S Column 5 mL	Strong Cation	732-4740
Foresight Nuvia HR-S Column 1 mL	Strong Cation	732-4723
Foresight Nuvia HR-S Column 5 mL	Strong Cation	732-4743
Foresight Nuvia Q Column 1 mL	Strong Anion	732-4721
Foresight Nuvia Q Column 5 mL	Strong Anion	732-4741
Foresight Nuvia HP-Q Column, 1 mL	Strong Anion	12007020
Foresight Nuvia HP-Q Column, 5 mL	Strong Anion	12007021
Foresight Nuvia IMAC Column, 1 mL	Ni-Charged	12004038
Foresight Nuvia IMAC Column, 5 mL	Ni-Charged	12004037
Foresight Nuvia aPrime 4A column, 1 mL	Mixed Mode (Hydrophobic-Anionic)	12007392
Foresight Nuvia aPrime 4A column, 5 mL	Mixed Mode (Hydrophobic-Anionic)	12007393
Foresight Nuvia cPrime Column, 1 mL	Mixed Mode (Hydrophobic-Weak Cationic)	732-4722
Foresight Nuvia cPrime Column, 5 mL	Mixed Mode (Hydrophobic-Weak Cationic)	732-4742

Table 6. Foresight Chromatography Media Columns, continued

Product	Mode	Catalog No.
Foresight Nuvia wPrime 2A Column, 1 mL	Mixed Mode (Hydrophobic-Weak Cationic)	12020456
Foresight Nuvia wPrime 2A Column, 5 mL	Mixed Mode (Hydrophobic-Weak Anionic)	12020484
Foresight UNOsphere S Column, 1 mL	Strong Cation	732-4730
Foresight UNOsphere S Column, 5 mL	Strong Cation	732-4750
Foresight UNOsphere Q Column, 1 mL	Strong Anion	732-4732
Foresight UNOsphere Q Column, 5 mL	Strong Anion	732-4752
Foresight CHT Type I, 40 µm Column, 1 mL	Mixed Mode Metal Affinity-Cationic	732-4735
Foresight CHT Type I, 40 µm Column, 5 mL	Mixed Mode Metal Affinity-Cationic	732-4755
Foresight CHT Type II, 40 µm Column, 1 mL	Mixed Mode Metal Affinity-Cationic	732-4736
Foresight CHT Type II, 40 µm Column, 5 mL	Mixed Mode Metal Affinity-Cationic	732-4756
Foresight CHT XT, 40 µm Column, 1 mL	Mixed Mode Metal Affinity-Cationic	12003150
Foresight CHT XT, 40 µm Column, 5 mL	Mixed Mode Metal Affinity-Cationic	12003149

Table 7. Foresight Chromatography Media Filter Plates

Product	Mode	Catalog No.
Foresight Nuvia S Plates, 20 µL	Strong Cation	732-4701
Foresight Nuvia HR-S Plates, 20 µL	Strong Cation	732-4707
Foresight Nuvia Q Plates, 20 µL	Strong Anion	732-4703

Table 7. Foresight Chromatography Media Filter Plates, continued

Product	Mode	Catalog No.
Foresight Nuvia HP-Q Plates, 20 µL	Strong Anion	12006908
Foresight Nuvia IMAC Plates, 20 µL	Ni-Charged	12004035
Foresight Nuvia aPrime 4A Plates, 20 µL	Mixed Mode (Hydrophobic-Anionic)	12007411
Foresight Nuvia cPrime Plates, 20 µL	Mixed Mode (Hydrophobic-Weak Cationic)	732-4705
Foresight Nuvia wPrime 2A Plates, 20 µL	Mixed Mode (Hydrophobic-Weak Anionic)	12020482
Foresight UNOsphere S Plates, 20 µL	Strong Cation	732-4710
Foresight UNOsphere Q Plates, 20 µL	Strong Anion	732-4714
Foresight CHT Type I 40 µm Plates, 20 µL	Mixed Mode (Metal Affinity-Cationic)	732-4716
Foresight CHT Type II 40 µm Plates, 20 µL	Mixed Mode (Metal Affinity-Cationic)	732-4718
Foresight CHT XT, 40 µm, 20 µL	Mixed Mode (Metal Affinity-Cationic)	12003151

Table 8. Foresight RoboColumns Units

Product	Mode	Catalog No.
Foresight Nuvia S RoboColumn, 200 µL	Strong Cation	732-4801
Foresight Nuvia S RoboColumn, 600 µL	Strong Cation	732-4802
Foresight Nuvia HR-S RoboColumn, 200 µL	Strong Cation	732-4831
Foresight Nuvia HR-S RoboColumn, 600 µL	Strong Cation	732-4832
Foresight Nuvia Q RoboColumn, 200 µL	Strong Anion	732-4804
Foresight Nuvia Q RoboColumn, 600 µL	Strong Anion	732-4805
Foresight Nuvia HP-Q RoboColumn, 200 µL	Strong Anion	12007013

Table 8. Foresight RoboColumns Units, continued

Product	Mode	Catalog No.
Foresight Nuvia HP-Q RoboColumn, 600 µL	Strong Anion	12007014
Foresight Nuvia IMAC RoboColumn, 200 µL	Ni-Charged	12004051
Foresight Nuvia IMAC RoboColumn, 600 µL	Ni-Charged	12004052
Foresight Nuvia aPrime 4A RoboColumn, 200 µL	Mixed Mode (Hydrophobic-Anionic)	12007394
Foresight Nuvia aPrime 4A RoboColumn, 600 µL	Mixed Mode (Hydrophobic-Anionic)	12007395
Foresight Nuvia cPrime RoboColumn, 200 µL	Mixed Mode (Hydrophobic-Weak Cationic)	732-4807
Foresight Nuvia cPrime RoboColumn, 600 µL	Mixed Mode (Hydrophobic-Weak Cationic)	732-4808
Foresight Nuvia wPrime 2A RoboColumn, 200 µL	Mixed Mode (Hydrophobic-Weak Anionic)	12020485
Foresight Nuvia wPrime 2A RoboColumn, 600 µL	Mixed Mode (Hydrophobic-Weak Anionic)	12020417
Foresight UNOsphere S RoboColumn, 200 µL	Strong Cation	732-4813
Foresight UNOsphere S RoboColumn, 600 µL	Strong Cation	732-4814
Foresight UNOsphere Q RoboColumn, 200 µL	Strong Anion	732-4819
Foresight UNOsphere Q RoboColumn, 600 µL	Strong Anion	732-4820
Foresight CHT Type I, 40 µm RoboColumn, 200 µL	Mixed Mode Metal Affinity-Cationic	732-4822
Foresight CHT Type I, 40 µm RoboColumn, 600 µL	Mixed Mode Metal Affinity-Cationic	732-4823
Foresight CHT Type II, 40 µm RoboColumn, 200 µL	Mixed Mode Metal Affinity-Cationic	732-4825
Foresight CHT Type II, 40 µm RoboColumn, 600 µL	Mixed Mode Metal Affinity-Cationic	732-4826

Table 8. Foresight RoboColumns Units, continued

Product	Mode	Catalog No.
Foresight CHT XT, 40 µm RoboColumn, 200 µL	Mixed Mode Metal Affinity-Cationic	12003152
Foresight CHT XT, 40 µm RoboColumn, 600 µL	Mixed Mode Metal Affinity-Cationic	12003148

Table 9. Bio-Rad Column Fitting Part Numbers

Catalog No.	Description
731-8226	3.2 mm Barb to Male Luer, pkg of 25
731-8225	1.6 mm Barb to Male Luer, pkg of 25
731-8224	0.8 mm Barb to Male Luer, pkg of 25
750-0564	HPLC Column to Biologic System Adaptors
760-1311	Long Fingertight Fittings, 10-32 x 1.03"



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